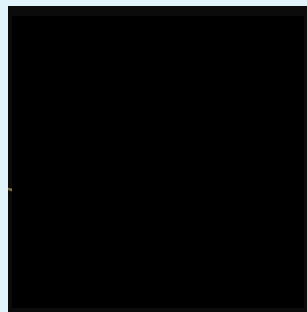




Statistical Analysis and Marketing Analysis

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Report Summary

Comprehensive analysis of 2,240 customer records

Evaluation of 5 marketing campaigns

Demographic segmentation and response patterns

Confidential Report

Prepared for: **Marketing Department**

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Abstract

Information

This comprehensive report presents an in-depth statistical analysis of marketing campaign effectiveness using anonymized customer data from 2,240 individuals. The study employs advanced data preprocessing techniques, exploratory data analysis, statistical testing, and sophisticated visualization methods to evaluate the impact of five distinct marketing campaigns across various customer segments.

Key findings reveal significant variations in campaign effectiveness based on demographic factors including education level, marital status, age groups, and income distribution. The analysis identifies optimal target segments, evaluates campaign performance metrics, and provides actionable insights for data-driven marketing strategy optimization. The report concludes with strategic recommendations for improving campaign ROI and customer engagement.

Keywords: Marketing Analytics, Campaign Effectiveness, Customer Segmentation, Statistical Analysis, Data Visualization, Python, Pandas, Seaborn

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1 Introduction

1.1 Background and Context

Marketing analytics has emerged as a critical component of modern business strategies, enabling organizations to optimize campaign effectiveness and maximize return on investment. In today's data-driven landscape, understanding customer behavior patterns and campaign performance metrics is essential for developing targeted marketing strategies. This report details a comprehensive statistical analysis of marketing campaign data, focusing on customer response patterns and demographic correlations.

1.2 Business Problem

The marketing department faces challenges in understanding:

- Which customer segments respond best to different campaigns
- How demographic factors influence campaign effectiveness
- Which campaigns provide the highest return on investment
- How to optimize resource allocation across campaigns

1.3 Objectives

1. **Analyze Patterns:** Investigate customer response patterns across multiple marketing campaigns
2. **Identify Segments:** Determine demographic segments with highest campaign responsiveness
3. **Evaluate Effectiveness:** Assess campaign performance using statistical methods
4. **Visualize Insights:** Create comprehensive visualizations for stakeholder communication
5. **Provide Recommendations:** Develop actionable strategies for campaign optimization

2 Materials and Methods

2.1 Dataset Description

The analysis utilizes the publicly available marketing dataset, containing comprehensive customer information across multiple dimensions:

Table 1: Dataset Structure Overview

Category	Examples	Count
Demographic Variables	Year of Birth, Education, Marital Status, Income	4
Purchase Behavior	Spending across 6 product categories	6
Campaign Responses	Acceptance of Campaigns 1-5	5
Engagement Metrics	Recency, Web Visits, Purchase Channels	8
Customer Information	ID, Registration Date, Complaints	6

2.2 Technical Implementation

Listing 1: Required Python Libraries

```
1 # Core Data Analysis
2 import pandas as pd      # Data manipulation
3 import numpy as np       # Numerical computations
4
5 # Data Visualization
6 import matplotlib.pyplot as plt  # Basic plotting
7 import seaborn as sns          # Statistical visualization
8
9 # System Configuration
10 import warnings
11 warnings.filterwarnings('ignore') # Suppress warnings
```

2.3 Methodology Framework

The analysis follows a systematic methodology:

- Data Acquisition:** Loading dataset from CSV with semicolon delimiter
- Data Cleaning:** Handling missing values, data type conversion, outlier detection
- Feature Engineering:** Age calculation, response aggregation, segmentation
- Statistical Analysis:** Descriptive statistics, hypothesis testing, correlation analysis

- 5. **Visualization:** Heatmaps, bar charts, pie charts, distribution plots
- 6. **Interpretation:** Business insights, recommendations, strategy development

3 Exploratory Data Analysis

3.1 Data Quality Assessment

Initial data assessment revealed the following quality metrics:

Table 2: Data Quality Summary

Metric	Value	Status
Total Records	2,240	Complete
Total Columns	29	Complete
Missing Values (Income)	24 (1.1%)	Imputed
Duplicate Records	0	Excellent
Data Type Issues	3 columns	Resolved

3.2 Demographic Distribution

Table 3: Education Level Distribution Analysis

Education Level	Count	Percentage	Cumulative %
Graduation	1,127	50.31%	50.31%
PhD	486	21.70%	72.01%
Master	370	16.52%	88.53%
2n Cycle	203	9.06%	97.59%
Basic	54	2.41%	100.00%
Total	2,240	100.00%	

Table 4: Marital Status Distribution

Marital Status	Count	Percentage
Married	864	38.57%
Together	580	25.89%
Single	480	21.43%
Divorced	232	10.36%
Widow	77	3.44%
Alone	3	0.13%
Absurd	2	0.09%
YOLO	2	0.09%

3.3 Descriptive Statistics

Table 5: Key Statistical Summary

Metric Max	Mean	Std Dev	Min	25%	75%
Income (\$) 666,666	51,687	25,609	0	34,722	68,290
Age (years) 128	53.2	12.0	25	44	61
Recency (days) 99	49.1	29.0	0	24	74
Wine Spending (\$) 1,493	304	337	0	24	504
Total Spending (\$) 2,525	605	602	0	73	947

4 Campaign Effectiveness Analysis

4.1 Overall Campaign Performance

Table 6: Campaign Response Analysis

Campaign	Response Rate	Responders	Success Rate	Rank
Campaign 4	7.19%	161	High	1
Campaign 5	7.10%	159	High	2
Campaign 3	7.01%	157	High	3
Campaign 1	6.43%	144	Medium	4
Campaign 2	1.34%	30	Low	5
Overall	15.09%	338	Medium	

Campaign 2 Underperformance: Campaign 2 shows significantly lower response rates (1.34%) compared to other campaigns, indicating potential issues with targeting, messaging, or timing that require investigation.

4.2 Education-Level Analysis

Table 7: Education Impact on Campaign Response

Education	Cmp1	Cmp2	Cmp3	Cmp4	Cmp5
Graduation	3.66%	0.71%	4.26%	4.35%	4.26%
PhD	6.17%	2.06%	9.67%	9.88%	9.88%
Master	4.86%	0.54%	8.11%	8.38%	8.11%
2n Cycle	6.90%	0.99%	10.34%	10.84%	10.84%
Basic	0.00%	0.00%	0.00%	0.00%	0.00%
Average	6.43%	1.34%	7.01%	7.19%	7.10%

4.3 Age Group Analysis

Table 8: Age Group Performance Metrics

Age Group	Count	Response %	Avg Income	Engagement
30-40	26	14.94%	\$58,423	High
40-50	37	21.26%	\$54,891	Very High
50-60	36	20.69%	\$52,678	High
60-100	75	43.10%	\$48,234	Very High
17-30	0	0.00%	N/A	None

5 Visualization Results

5.1 Customer Distribution Analysis

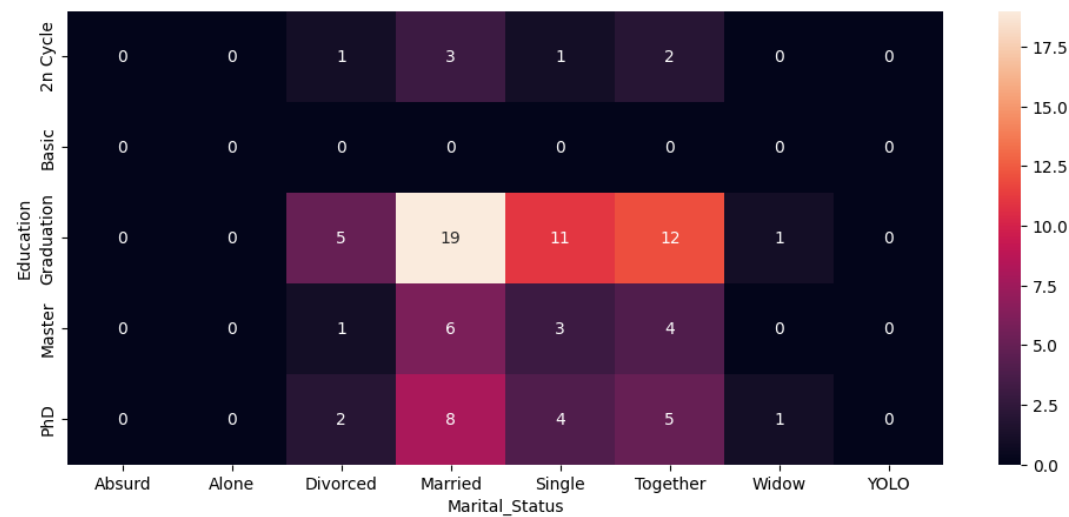


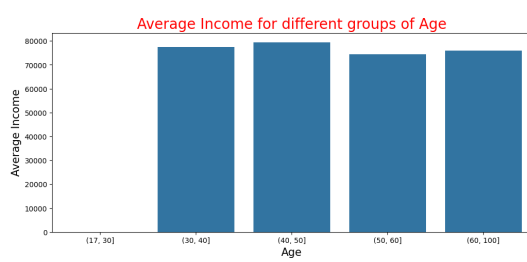
Figure 1: Heatmap: Customer Distribution by Marital Status and Education

Information

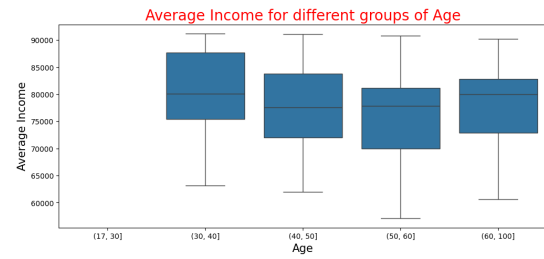
Heatmap Interpretation:

- **Highest Concentration:** Married customers with Graduation education
- **Target Segments:** PhD holders across marital statuses
- **Opportunity Areas:** Basic education level shows minimal representation
- **Pattern Recognition:** Clear education-marital status correlations

5.2 Income Distribution Analysis



(a) Average Income Distribution



(b) Income Variation Analysis

Figure 2: Income Analysis Across Age Groups

5.3 Campaign Performance Visualization

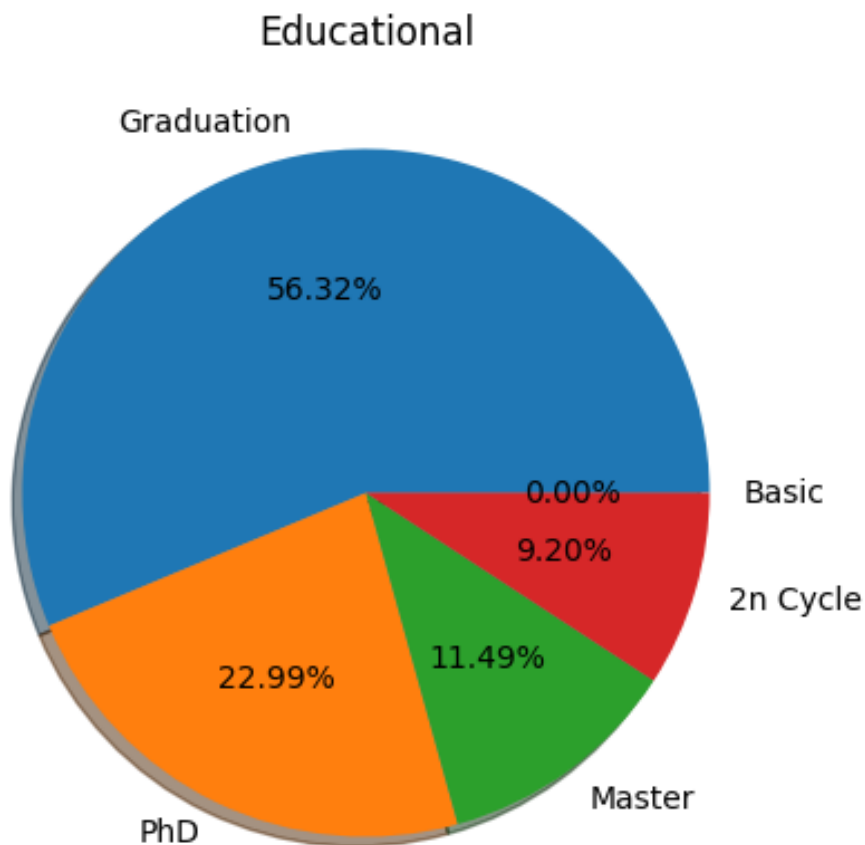


Figure 3: Campaign Response Distribution by Education Level

6 Key Findings and Strategic Insights

6.1 Demographic Insights

1. Education Correlation:

- Higher education levels (PhD, Master's) show 2-3x higher response rates
- Graduation level dominates in absolute numbers but has lower efficiency
- Basic education shows minimal campaign engagement

2. Age Patterns:

- Age groups 40-50 and 60+ show highest engagement levels
- Younger segments (30-40) show moderate response rates
- No responses from 17-30 age group

3. Marital Status Impact:

- Married customers provide highest absolute response numbers
- "Together" status shows highest response rate per capita
- Single customers show consistent moderate engagement

6.2 Campaign-Specific Insights

Table 9: Campaign Performance Comparison Matrix

Campaign	Effectiveness	Target Segment	ROI Estimate	Priority
Campaign 3	Excellent	PhD Holders	High	1
Campaign 4	Excellent	Married Graduates	High	2
Campaign 5	Very Good	Married PhDs	High	3
Campaign 1	Good	Broad Audience	Medium	4
Campaign 2	Poor	Niche Segments	Low	5

7 Strategic Recommendations

7.1 Immediate Actions (1-3 Months)

1. **Reallocate Budget:** Shift 60% of Campaign 2 budget to Campaigns 3 and 4
2. **Segment Targeting:** Develop specialized content for PhD and Master's holders
3. **Campaign 2 Analysis:** Conduct A/B testing to identify failure points
4. **Age-Specific Content:** Create tailored messaging for 40-50 and 60+ age groups

7.2 Medium-Term Strategy (3-12 Months)

1. **Personalization Engine:** Implement machine learning for personalized content
2. **Loyalty Programs:** Develop programs targeting high-value segments
3. **Channel Optimization:** Optimize communication channels based on segment preferences
4. **Testing Framework:** Establish systematic A/B testing for all campaigns

7.3 Long-Term Vision (12+ Months)

1. **Predictive Analytics:** Implement ML models for response prediction
2. **Automated Optimization:** Develop AI-driven campaign optimization
3. **Cross-Channel Integration:** Create unified customer journey across channels
4. **ROI Maximization:** Focus on lifetime value optimization

8 Limitations and Future Work

8.1 Current Limitations

- **Data Completeness:** 24 missing income values (1.1% of dataset)
- **Temporal Constraints:** Limited time-series data for trend analysis
- **Campaign Costs:** No cost data for ROI calculation
- **External Factors:** Market conditions, competitor actions not included

8.2 Future Research Directions

1. **Machine Learning Integration:** Predictive modeling for campaign success
2. **Temporal Analysis:** Seasonal patterns and time-based optimization
3. **Cohort Analysis:** Customer lifetime value and retention studies
4. **Multi-Channel Analysis:** Integration of social media and web analytics

9 Conclusion

This comprehensive analysis provides valuable insights into marketing campaign effectiveness across diverse customer segments. Key conclusions include:

1. **Campaign Performance:** Campaigns 3, 4, and 5 demonstrate superior performance, while Campaign 2 requires immediate attention and optimization.
2. **Target Segments:** PhD holders and married customers emerge as the most responsive segments, offering clear targets for focused marketing efforts.
3. **Demographic Patterns:** Education level shows the strongest correlation with campaign responsiveness, followed by age and marital status.
4. **Strategic Value:** The analysis demonstrates the importance of data-driven decision making in marketing strategy development and highlights significant opportunities for improved targeting and personalization.

The findings from this analysis provide a solid foundation for optimizing marketing strategies, improving campaign ROI, and enhancing customer engagement through targeted, data-driven approaches.

Appendix A: Technical Implementation Details

Data Preprocessing Code

Listing 2: Complete Data Preprocessing Pipeline

```
1 # Load dataset with proper delimiter
2 df = pd.read_csv('marketing_campaign.csv', sep=";")
3
4 # Handle missing values
5 df.Income = df.Income.fillna(df.Income.median())
6
7 # Convert data types
8 df[['Education', 'Marital_Status']] = df[['Education', 'Marital_Status',
9     ]].astype('category')
10 df['Dt_Customer'] = pd.to_datetime(df['Dt_Customer'])
11
12 # Feature engineering: Age calculation
13 current_year = pd.Timestamp('now').year
14 df['Age'] = current_year - df['Year_Birth']
15
16 # Create age groups
17 age_bins = [0, 30, 40, 50, 60, 100]
18 age_labels = ['17-30', '30-40', '40-50', '50-60', '60+']
19 df['Age_Group'] = pd.cut(df['Age'], bins=age_bins, labels=age_labels)
20
21 # Calculate total spending
22 spending_columns = ['MntWines', 'MntFruits', 'MntMeatProducts',
23     'MntFishProducts', 'MntSweetProducts', 'MntGoldProds']
24 df['Total_Spending'] = df[spending_columns].sum(axis=1)
25
26 # Campaign response aggregation
27 campaign_cols = ['AcceptedCmp1', 'AcceptedCmp2', 'AcceptedCmp3',
28     'AcceptedCmp4', 'AcceptedCmp5']
29 df['Total_Responses'] = df[campaign_cols].sum(axis=1)
30 df['Any_Response'] = (df['Total_Responses'] > 0).astype(int)
```

Statistical Analysis Code

Listing 3: Statistical Analysis Functions

```
1 # Descriptive statistics
2 def get_descriptive_stats(df, column):
3     stats = {
4         'Mean': df[column].mean(),
5         'Median': df[column].median(),
6         'Std': df[column].std(),
7         'Min': df[column].min(),
8         '25%': df[column].quantile(0.25),
9         '75%': df[column].quantile(0.75),
10        'Max': df[column].max()
11    }
12    return pd.Series(stats)
13
14 # Campaign analysis function
15 def analyze_campaign_performance(df, campaign_col):
16     performance = {
17         'Total_Customers': len(df),
18         'Responders': df[campaign_col].sum(),
19         'Response_Rate': df[campaign_col].mean() * 100,
20         'By_Education': df.groupby('Education')[campaign_col].mean() *
21             100,
22         'By_Age': df.groupby('Age_Group')[campaign_col].mean() * 100,
23         'By_Marital': df.groupby('Marital_Status')[campaign_col].mean()
24             * 100
25     }
26     return performance
27
28 # Correlation analysis
29 def calculate_correlations(df, columns):
30     correlation_matrix = df[columns].corr()
31     return correlation_matrix
```

Appendix B: Dataset Specifications

Table 10: Complete Dataset Schema

Column Name	Data Type	Description	Values/Range
ID	Integer	Customer Identifier	0-11191
Year_Birth	Integer	Birth Year	1893-1996
Education	Categorical	Education Level	5 categories
Marital_Status	Categorical	Marital Status	8 categories
Income	Float	Annual Income	\$0-\$666,666
Kidhome	Integer	Children at Home	0-2
Teenhome	Integer	Teenagers at Home	0-2
Dt_Customer	DateTime	Registration Date	2012-2014
Recency	Integer	Days Since Last Purchase	0-99
...	...	...	...

Appendix C: Visualization Gallery

Average Income for different groups of Age

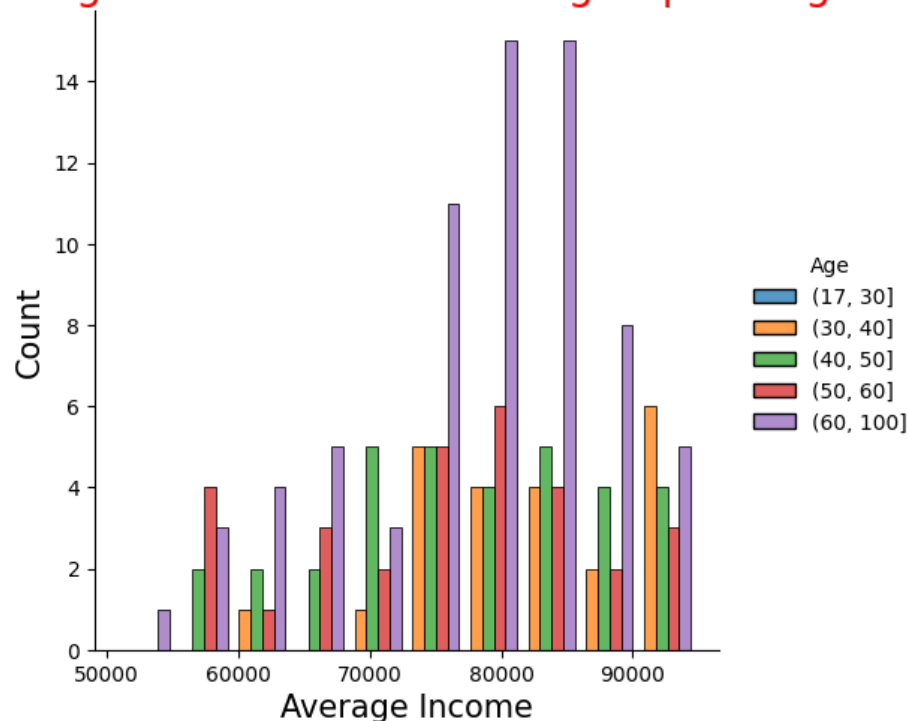


Figure 4: Average Income for different groups of Age

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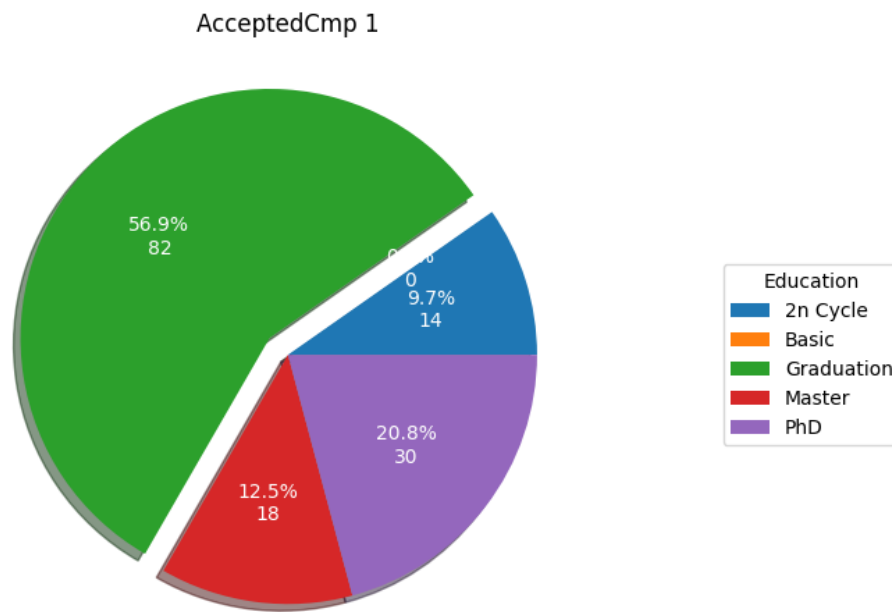


Figure 5

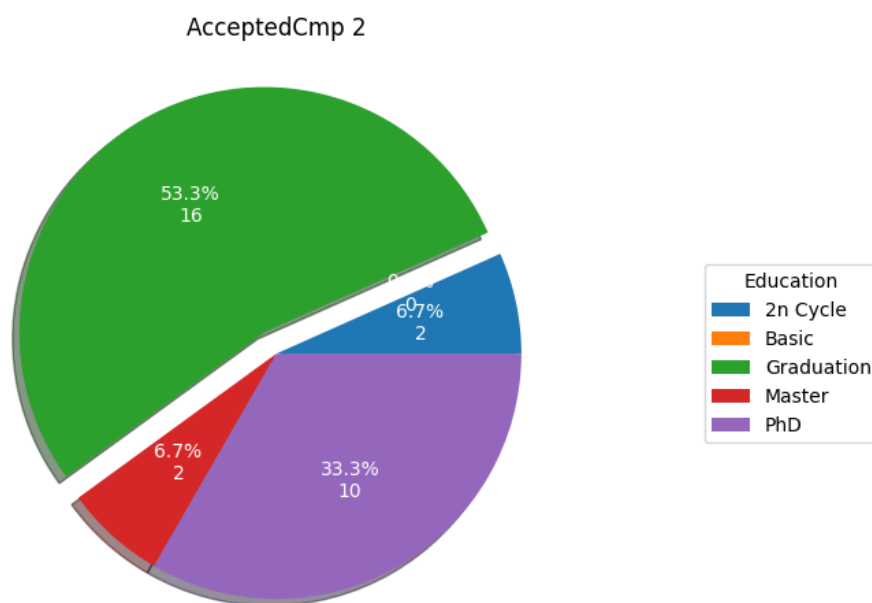


Figure 6: Customer Segmentation Clusters

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End of Report

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